Math Study Guide

May 7, 2026

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1 Topic Name

1.1 Definitions

Definition 1.1. Orbit:

How to find it:

Definition 1.2. Stabilizer:

How to find it:

Definition 1.3. Orbit-Stabilizer Theorem:

Definition 1.4.

Definition 1.5. Lagrange's Theorem:

The size of the group is equal to the number of cosets times the size of each coset.

$$| G |=| G/H | \cdot | H |$$

The size of the quotient group of a group G by a subgroup H is equal to the number of cosets of H in G.

Definition 1.6.

1.2 Key Theorems

Theorem 1.7. *Chinese Remainder Theorem:*

For coprime integers, you have the isomorphism:

$$Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_k} \cong Z_{n_1 n_2 \ldots n_k}$$

1.3 Examples

Example 1.8.

1.4 Common Proof Patterns

Lemma 1.9.

Proof. □

1.5 Exercises

Exercise 1.10.